

CHAPTER – 1

ACCIDENTS IN INDIA

Introduction

This is the fourth edition of the report after the comprehensive revision of the proformae for data collection of 'Accidental Deaths & Suicides in India' completed in 2014 in consultation with States/UTs Police and others stakeholders.

An accident is an unplanned event that results in injury, deaths, damage to property or some other losses. For statistical purpose, Accidental Deaths have been classified as deaths that could be due to forces/factors of

the nature which have been termed as 'Accidental Deaths due to Forces of Nature' or could be due to deliberate or negligent conduct of human beings, which is termed as 'Other causes of Accidental Deaths'.

The rate of accidental deaths (per lakh of population) has decreased in 2017 as compared to 2016. The total number of accidental deaths in 2017 was 3,96,584 [Table-1(A)]. The incidence and rate of accidental deaths during 2007-2017 is presented in Table-1.1.

Table – 1 (A)
Percentage Change in Population, Number and Rate of Accidental Deaths over the Corresponding Previous Year during 2013 to 2017

Year	Mid-Year Projected Population* (In Lakh+)	Percentage Change in Population over the Previous Year	Percentage Change in Accidental Deaths over the Previous Year	Total Number of Accidental Deaths	Rate of Accidental Deaths	Percentage Change in Rate of Accidental Deaths over the Previous Year
(1)	(2)	(3)	(4)	(5)	(6)	(7)
2013	12287.9	1.3	1.4	400517	32.6	0.0
2014	12440.4	1.2	12.8	451757	36.3	11.3
2015	12591.1	1.2	-8.5	413457	32.8	-9.6
2016	12739.9	1.2	1.2	418221	32.8	0.0
2017	13091.6#	2.8	-5.2	396584	30.3	-7.6

* Source: The Registrar General of India.

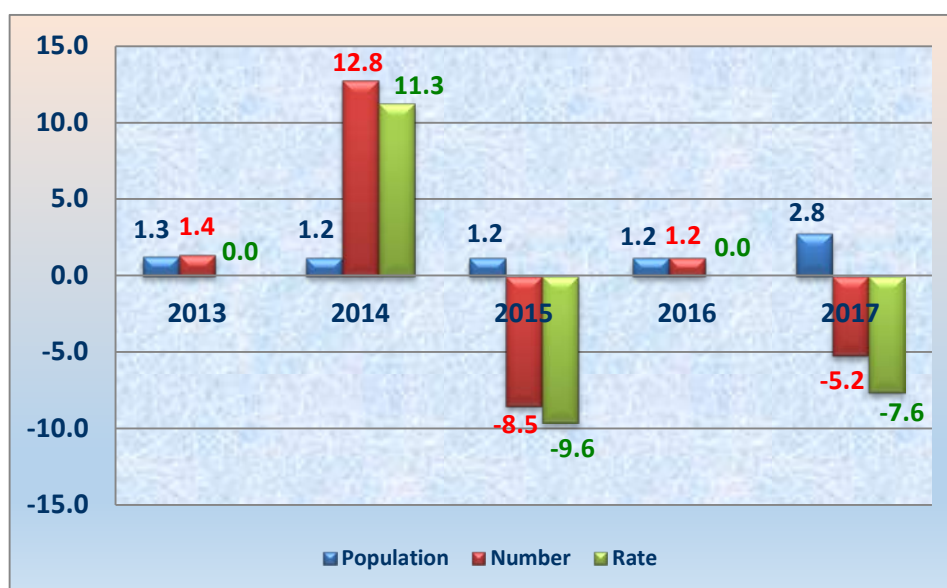
Source: Report of the Technical Group on Population Projections (November, 2019), National Commission on Population, Ministry of Health & Family Welfare.

+ One Lakh = 0.1 Million = 100,000

• As per data provided by States/UTs.

Figure 1.1

Percentage Change in Population, Number and Rate of Accidental Deaths during 2013 – 2017 (Over Corresponding Previous Year)



- As per data provided by States/UTs.

Out of 3,96,584 accidental deaths, 7,143 (1.8%) deaths were due to forces of nature and 3,89,441 (98.2%) deaths were due to other causes.

There was a decline of 17.7% (from 8,684 deaths in 2016 to 7,143 deaths in 2017) in deaths due to forces of nature and 4.9% (from 4,09,537 deaths in 2016 to 3,89,441 deaths in 2017) in deaths due to other causes during 2017 as compared to 2016 [Table-1(B)].

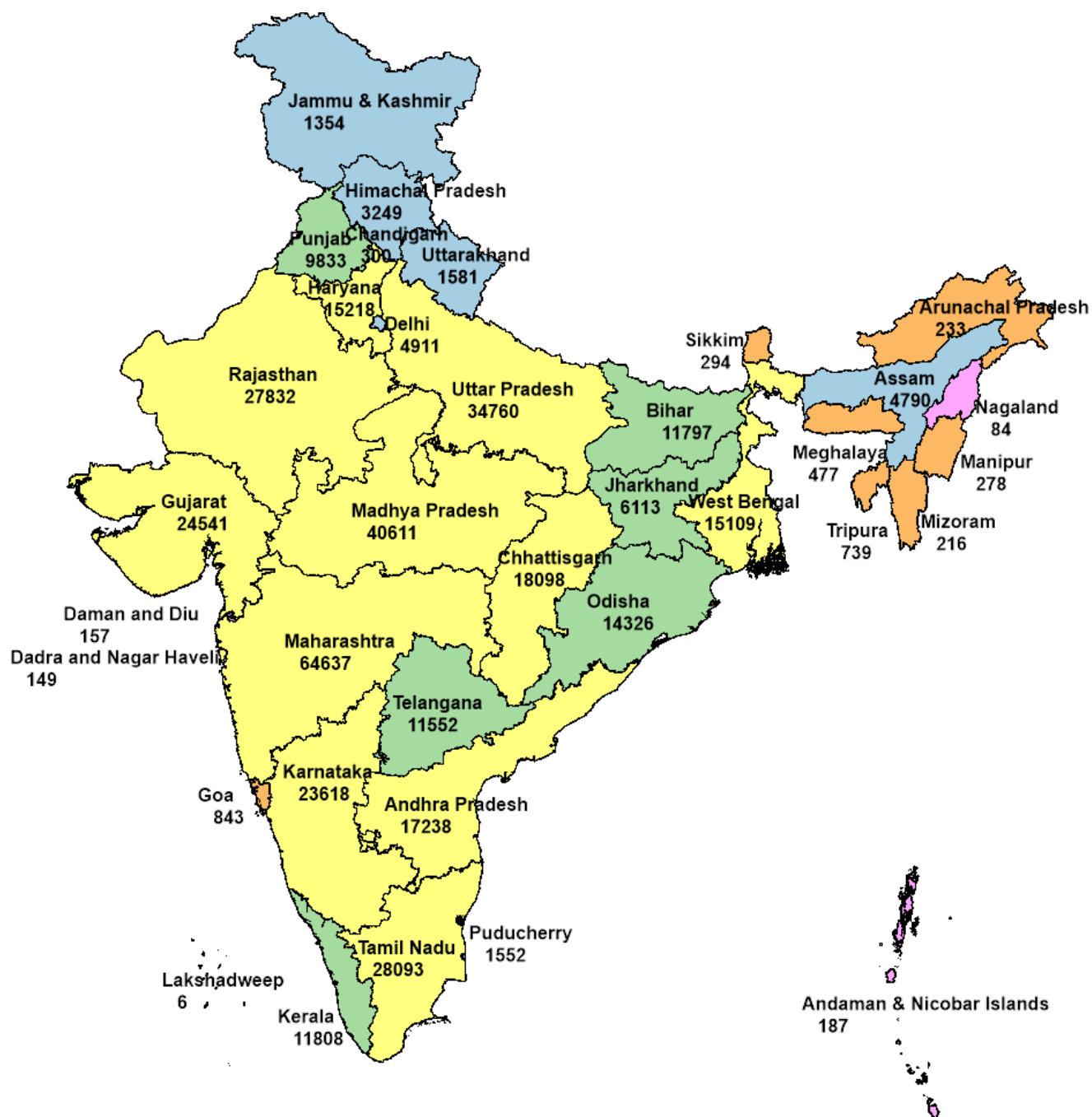
Table — 1 (B)

Number of Accidental Deaths due to Forces of Nature & Other Causes (2013 to 2017)

Year	Forces of Nature		Other Causes		Total Accidental Deaths	
	Number of Deaths	% change over previous year	Number of Deaths	% change over previous year	Number of Deaths	% change over previous year
(1)	(2)	(3)	(4)	(5)	(6)	(7)
2013	22,759	-0.9	3,77,758	1.5	4,00,517	1.4
2014	20,201	-11.2	4,31,556	14.2	4,51,757	12.8
2015	10,510	-48.0	4,02,947	-6.6	4,13,457	-8.5
2016	8,684	-17.4	4,09,537	1.6	4,18,221	1.2
2017	7,143	-17.7	3,89,441	-4.9	3,96,584	-5.2

- As per data provided by States/UTs.

STATE/UT - WISE ACCIDENTAL DEATHS DURING 2017



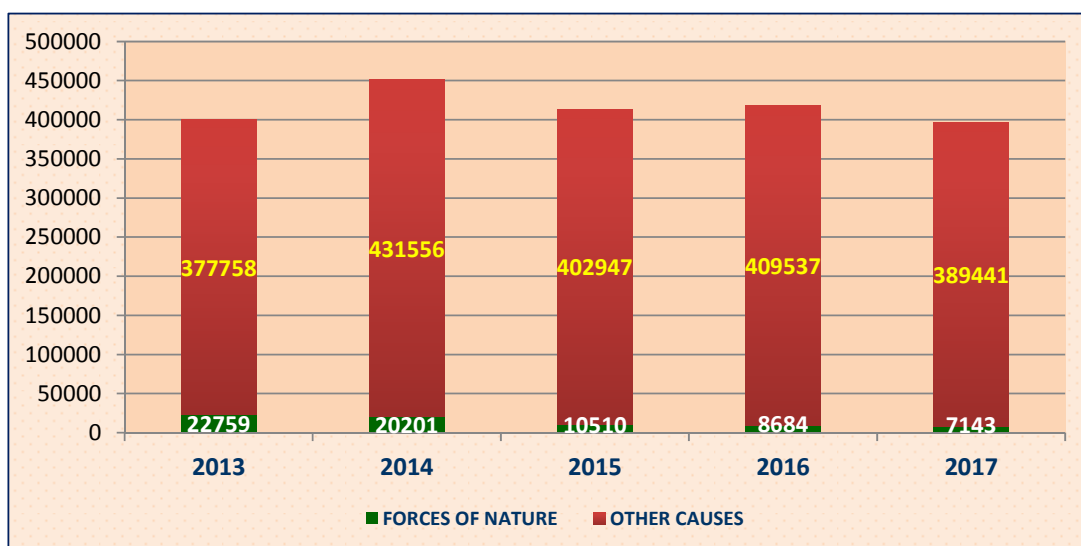
Accidental Deaths

	upto 200
	201 to 1000
	1001 to 5000
	5001 to 15000
	Above 15000

• As per data provided by States/UTs.

Map Powered by DevInfo, UNICEF

Figure 1.2
Number of Accidental Deaths due to Forces of Nature and Other Causes during 2013-2017



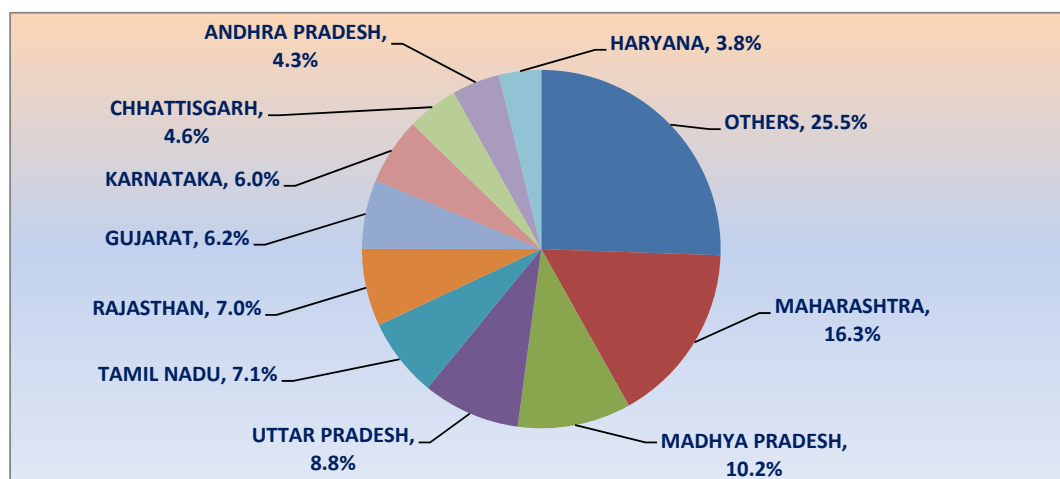
- As per data provided by States/UTs.

Number & Rate of Accidental Deaths in States, UTs and Cities

Out of 3,96,584 accidental deaths, like previous year, in 2017 also, Maharashtra with a population share of 9.2% has reported highest number of accidental deaths (64,637), contributing nearly one-sixth (16.3%) of total accidental deaths reported in the country. Uttar

Pradesh, the most populous State in the country with population share of 16.8% has accounted for 8.8% of the total accidental deaths in the country. The other States having higher percentage share in total accidental deaths were Madhya Pradesh (10.2%), Tamil Nadu (7.1%), Rajasthan (7.0%), Gujarat (6.2%), Karnataka (6.0%), Chhattisgarh (4.6%), Andhra Pradesh (4.3%) and Haryana (3.8%) during 2017 [Table-1.2].

Figure 1.3
Percentage Share of Major States in Accidental Deaths during 2017



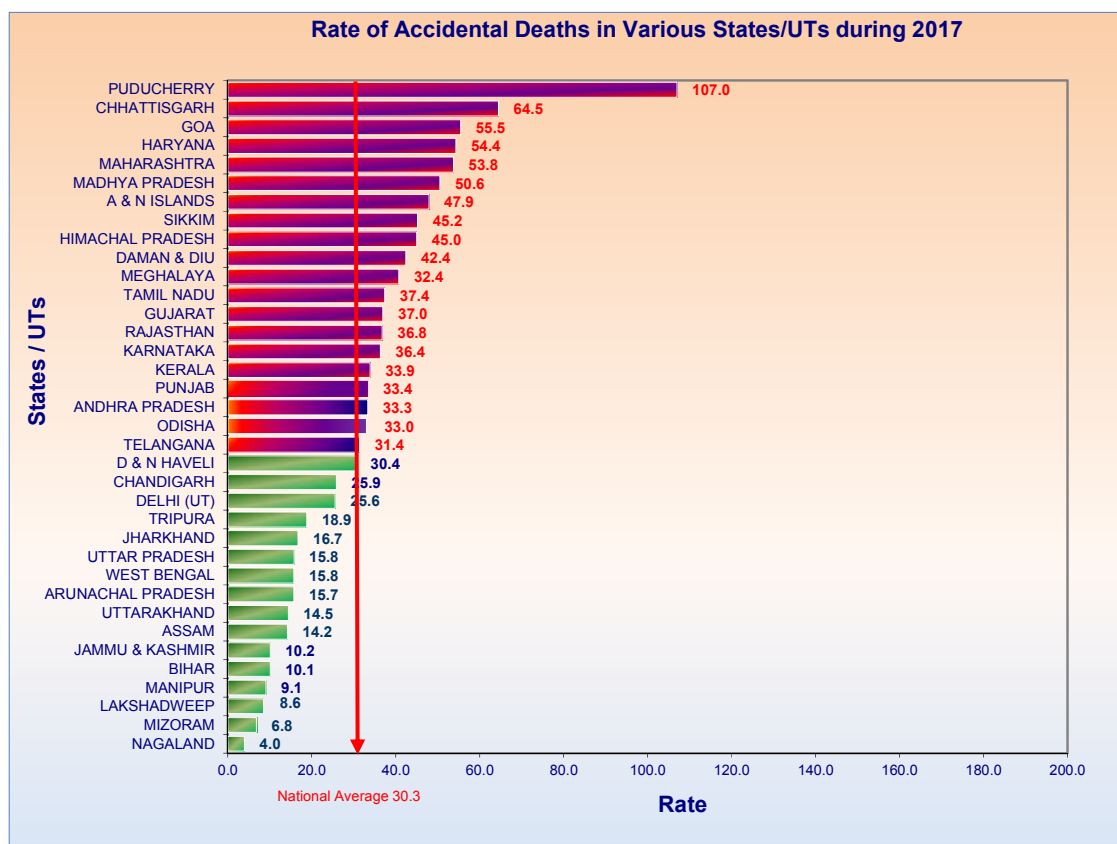
- As per data provided by States/UTs.

Other States/UTs include Arunachal Pradesh, Bihar, Assam, Chandigarh, Goa, West Bengal, Jammu & Kashmir, Jharkhand, Kerala, Manipur, Mizoram, Meghalaya, Nagaland, Odisha, Punjab, Sikkim, Telangana, Tripura, Uttarakhand, A & N Island, D&N Haveli, Daman & Diu, Delhi, Lakshadweep and Puducherry

The highest **rate of accidental deaths** was reported from Puducherry (107.0) followed by Chhattisgarh (64.5), Goa (55.5), Haryana (54.4), Maharashtra (53.8) and Madhya Pradesh (50.6) against the national average rate of 30.3. 20 out of 36 States/UTs have

reported higher rates of 'Accidental Deaths' as compared to All India Average of 30.3 deaths per one lakh of population at all India level [Figure– 1.4].

Figure 1.4



- As per data provided by States/UTs.

The number of accidental deaths has decreased by 5.2% at national level during 2017 as compared to 2016. The number of 'Accidental Deaths' due to forces of nature have decreased by 17.7% and deaths due to other causes have decreased by 4.9% during 2017 over the year 2016. [Table-1.3].

Puducherry has reported the highest percentage increase (85.2%) in total accidental deaths in the year 2017 as compared to 2016 (from 838 in 2016 to 1,552 in 2017), followed by D & N Haveli (25.2%), Odisha (16.6%), Meghalaya (13.3%) and Bihar

(11.6%). Lakshadweep has reported a decline of 50.0% in total accidental deaths in the year 2017 as compared to 2016 followed by Arunachal Pradesh (48.9%), Chandigarh (29.6%) and Delhi UT (25.4%) [Table-1.3].

Gender-wise analysis reveals that female and male constitute 19.7% and 80.3% of total victims respectively. Age group of most of the victims was between 30 to 45 years. This group of people has accounted for 30.9% of all persons killed in accidents in the country during the year 2017. A total of 39,932 senior citizens (60 years & above of age) have also got killed in various accidents during 2017 [Table-1.4].

STATE/UT - WISE RATE OF ACCIDENTAL DEATHS DURING 2017



Note:

Rate of Accidental Deaths means Number of Accidental Deaths per 1,00,000 population.

Accidental Death Rate All India Average 30.3

	upto 10
	11 to 20
	21 to 30
	31 to 50
	Above 50

Map Powered by DevInfo, UNICEF

- As per data provided by States/UTs.

Causes of Accidental Deaths

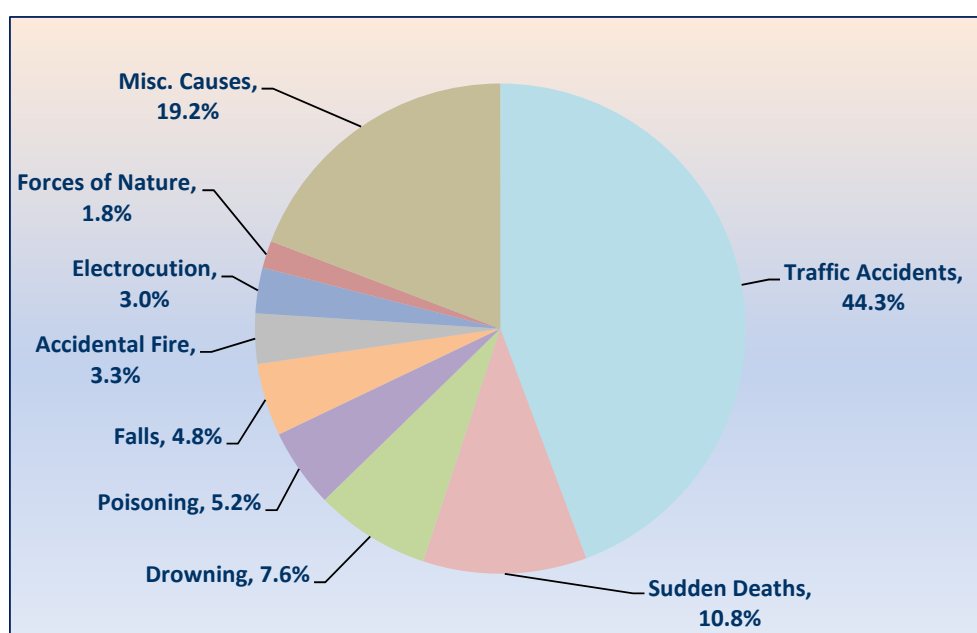
Accidental deaths have been grouped into two broad categories i.e. 'Deaths due to Forces of Nature' and 'Deaths due to Other Causes'. Classification of types and causes of accidental deaths has been done for statistical purpose only.

Under 'Causes Attributable to Forces of Nature', the deaths due to 'Avalanche', 'Exposure to Cold', 'Cyclone', 'Tornado', 'Landslide' and 'Torrential Rain' have increased during the year 2017 over the previous year 2016 whereas the deaths due to which *inter-alia* include "Earth quake", 'Epidemic', 'Flood', 'Heat/Sun Stroke', 'Lightning', and 'Forest Fire' have decreased in 2017 over 2016.

The deaths due to Other Causes such as 'Air Crash', 'Collapse of Structure', 'Drowning', 'Electrocution', 'Falls', 'Stampede', 'Sudden Deaths', 'Deaths due to Consumption of Illicit/Spurious Liquor', 'Killed by Animals', and 'Suffocation' has increased whereas the deaths due to 'Accidental Explosion', 'Factory/ Machine Accidents', 'Accidental Fire', 'Firearm', 'Mines or Quarry disaster', 'Traffic Accidents', 'Deaths of Women during Pregnancy', 'Poisoning' and 'Drug Overdose' has decreased in 2017 as compared to the previous year 2016.

Cause-wise details of accidental deaths, its share and percentage variation during 2017 over 2016 are presented in **Table-1(C)**.

Figure 1.5
Percentage Share of Various Major Causes of Accidental Deaths during 2017
(Forces of Nature and Other Causes)



- As per data provided by States/UTs.

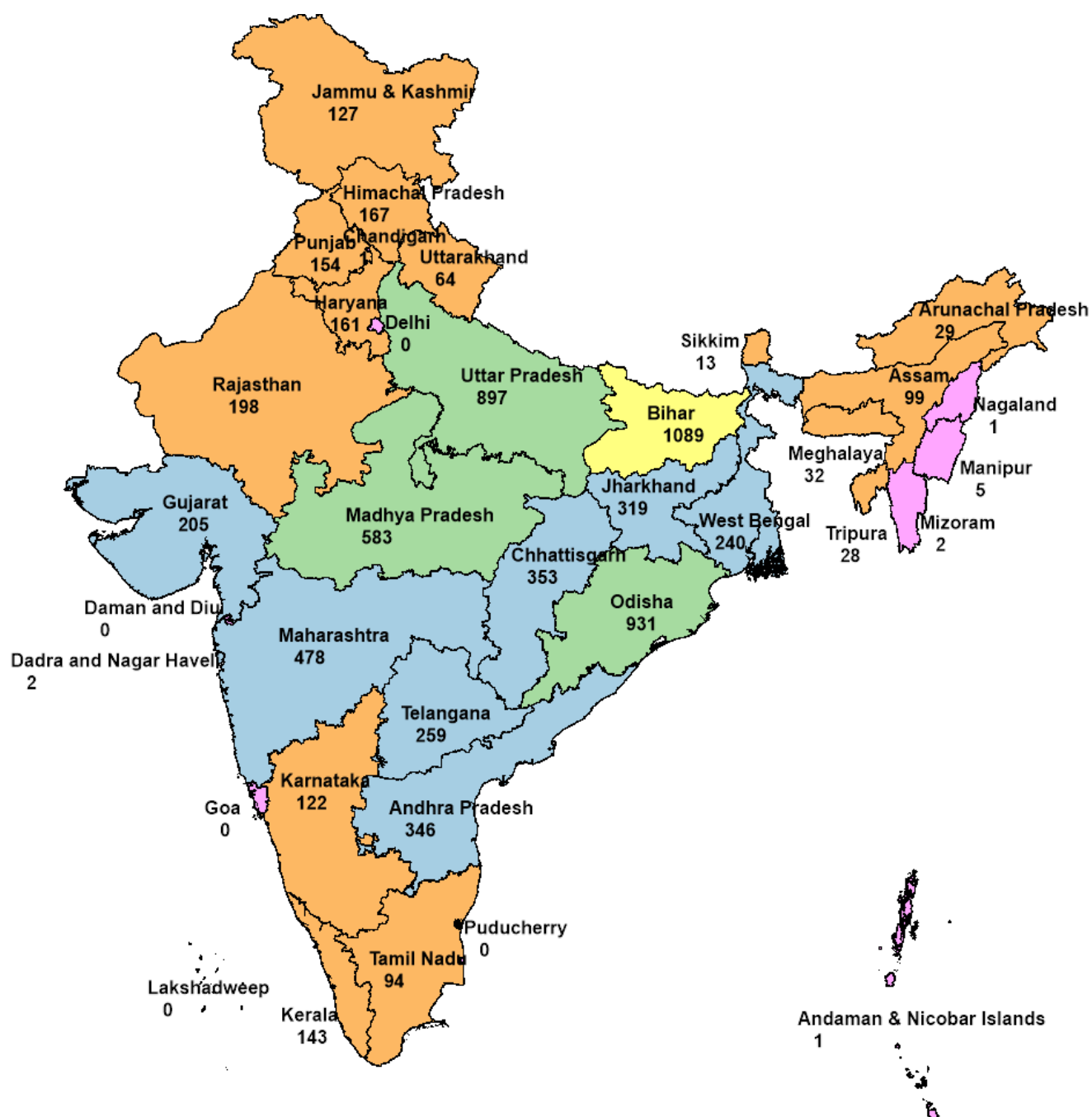
Misc. Causes include Collapse of Structure, Suffocation, Deaths of Women during Pregnancy, Killed by Animals, Deaths due to Consumption of Illicit/Poisonous Liquor, Accidental Explosion, Drug Overdose, Factory/Machine Accidents, Firearm, Mines or Quarry Disaster, Stampede, Air-Crash, Ship Accidents, Other than above & Unknown Causes.

Table-1(C)
Number and Share of Accidental Deaths due to Forces of Nature and Other Causes - 2016 & 2017

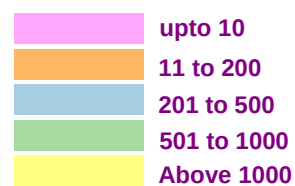
Sl. No.	Cause	2016		2017		% Variation during 2017 over 2016
		No.	% Share (w.r.t. All India)	No.	% Share (w.r.t. All India)	
(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. FORCES OF NATURE						
1	Avalanche	19	0.2	38	0.5	100.0
2	Exposure to Cold	687	7.9	738	10.3	7.4
3	Cyclone	15	0.2	133	1.9	786.7
4	Tornado	9	0.1	14	0.2	55.6
5	Tsunami	0	0.0	0	0.0	-
6	Earthquake	3	0.0	2	0.0	-33.3
7	Epidemic	36	0.4	3	0.0	-91.7
8	Flood	775	8.9	496	6.9	-36.0
9	Heat/Sun Stroke	1338	15.4	1127	15.8	-15.8
10	Landslide	210	2.4	231	3.2	10.0
11	Lightning	3315	38.2	2885	40.4	-13.0
12	Torrential Rain	143	1.6	259	3.6	81.1
13	Forest Fire	21	0.2	19	0.3	-9.5
14	Causes other than above	2113	24.3	1198	16.8	-43.3
15	Total (A)	8684	100.0	7143	100.0	-17.7
B. OTHER CAUSES						
1	Air-Crash	2	0.0	16	0.0	700.0
2	Ship Accidents	0	0.0	8	0.0	-
3	Collapse of Structure	1984	0.5	1997	0.5	0.7
4	Drowning	29721	7.3	30279	7.8	1.9
5	Electrocution	11126	2.7	12004	3.1	7.9
6	Accidental Explosion	852	0.2	729	0.2	-14.4
7	Falls	17278	4.2	19092	4.9	10.5
8	Factory/Machine Accidents	685	0.2	582	0.1	-15.0
9	Accidental Fire	16900	4.1	13159	3.4	-22.1
10	Firearm	614	0.1	540	0.1	-12.1
11	Mines or Quarry Disaster	130	0.0	93	0.0	-28.5
12	Traffic Accidents	177904	43.4	175586	45.1	-1.3
13	Stampede	45	0.0	49	0.0	8.9
14	Sudden Deaths	41628	10.2	42999	11.0	3.3
15	Deaths of Women during Pregnancy	1131	0.3	1086	0.3	-4.0
16	Deaths due to Consumption of Illicit/Poisonous Liquor	1054	0.3	1510	0.4	43.3
17	Killed by Animals	1063	0.3	1207	0.3	13.5
18	Poisoning	23094	5.6	20809	5.3	-9.9
19	Suffocation	1399	0.3	1725	0.4	23.3
20	Drug Overdose	778	0.2	745	0.2	-4.2
21	Causes Not Known	20979	5.1	12179	3.1	-41.9
22	Causes other than above	61170	14.9	53047	13.6	-13.3
	Total (B)	409537	100.0	389441	100.0	-4.9
	Grand Total (A+B)	418221	100.0	396584	100.0	-5.2

- As per data provided by States/UTs.

STATE/UT - WISE DEATHS DUE TO FORCES OF NATURE DURING 2017



Accidental Deaths



Map Powered by DevInfo, UNICEF

- As per data provided by States/UTs.

Accidental Deaths due to Forces of Nature

Cause - wise accidental deaths due to forces of nature are presented in **Table-1.5**. Out of 7,143 accidental deaths due to forces of nature, 40.4% deaths were due to 'Lightning' (2,885 deaths), 15.8% deaths due to 'Heat/Sun Stroke' (1,127 deaths), 10.3% deaths due to 'Exposure to Cold' (738 deaths) and 6.9% deaths due to 'Flood' (496 deaths) during the year 2017. 'Torrential Rain' and 'Landslide' caused 3.6% (259 deaths) and 3.2% (231 deaths) of total deaths due to forces of nature respectively during 2017.

The age group-wise distribution of accidental deaths due to forces of nature is given in **Table-1.5**. As per age-wise classification of deaths due to forces of nature, 7.5% of such victims were below 14 years of age (533 out of 7,143 deaths), 6.9% in the age-group of 14 years & above to below 18 years (492 deaths), 20.5% in the age-group of 18 years to below 30 years (1,465 deaths), 29.5% in the age-group of 30 years to below 45 years (2,104 deaths) and 24.7% victims were in age group 45 years to below 60 years (1,763 deaths) during 2017. Senior citizens (60 yrs. and above) constituted 11.0% (786 deaths) of the total such victims.

State/UT - wise and gender - wise distribution of accidental deaths in the country by forces of nature along with their share in total deaths is presented **Table-1.8**. 'Lightning' caused 100.0%, 100.0% and 99.2%, of total such deaths in A & N Islands (1 out of 1 death), D & N Haveli (2 out of 2 deaths) and Karnataka (121 out of 122 deaths) respectively. 'Heat/Sun Stroke' caused 69.5% (180 out of 259 deaths) of total accidental deaths due to forces of nature in Telangana. 100% (1 out of 1 death) of death in Chandigarh UT were due to 'Exposure to Cold'. Similarly, 100% (1 out of 1 deaths) of deaths in Nagaland was due to 'Cyclone'. 'Landslide' caused 61.5%, 60.0%, 50.0% and 40.1% of total deaths in Sikkim (8 out of 13), Manipur (3 out of 5 deaths), Mizoram (1 out of 2 deaths) and Himachal Pradesh (65 out of 162 deaths) respectively during 2017.

Majority of accidental deaths due to 'Flood' were reported in Bihar which accounted for 69.8% (346 out of 496 deaths) of total such deaths reported in the country during 2017.

Majority of deaths due to 'Torrential Rain' were reported in Bihar which accounted for 33.6% (87 out of 259 deaths) of total such deaths during 2017.

Other Causes of Accidental Deaths

It is generally accepted that accidents due to 'Other Causes' are preventive and can be reduced by effective safety measures and safety consciousness.

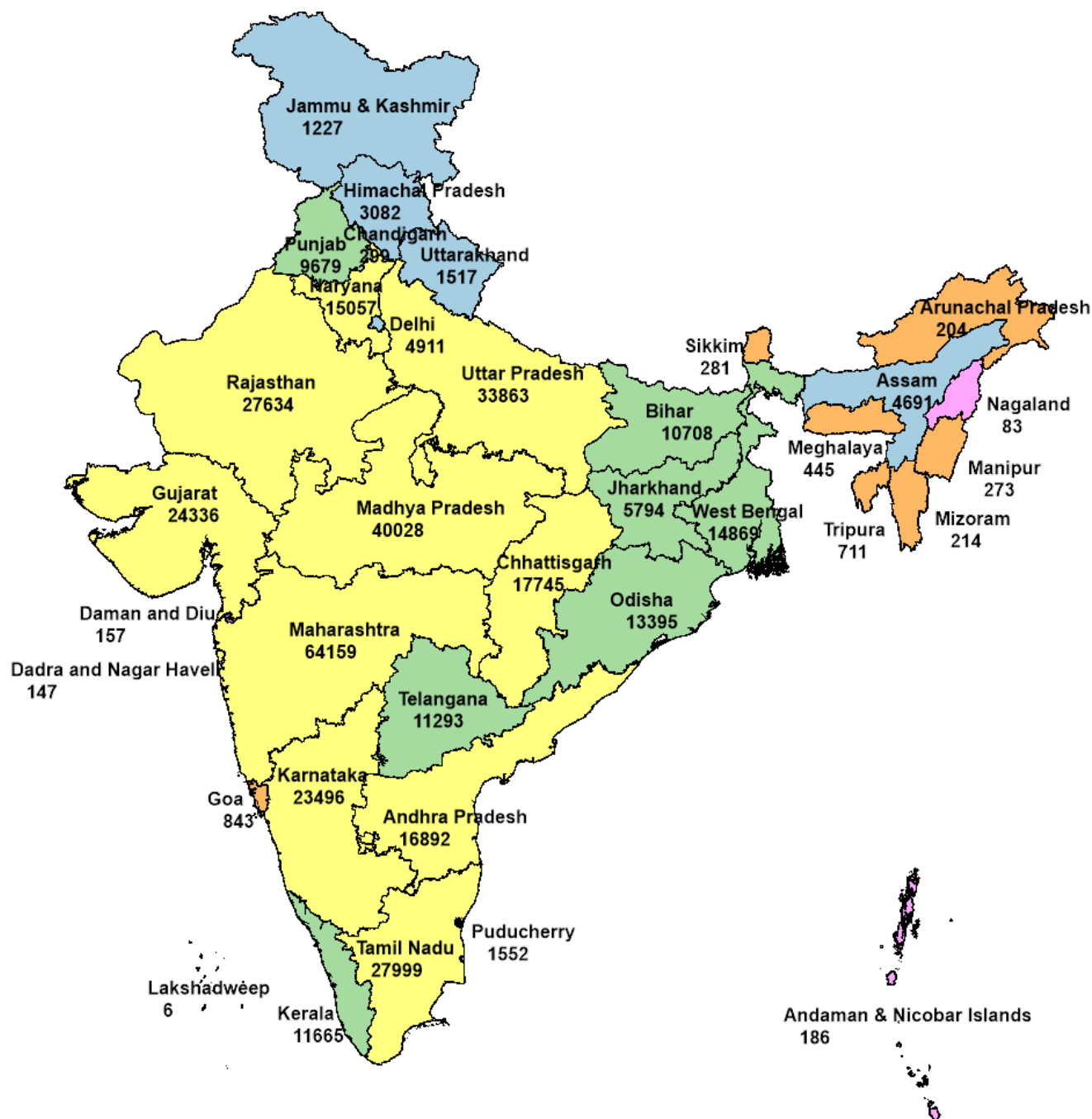
A total of 6,89,556 cases of Accidents due to 'Other Causes' were reported in the country which resulted in 3,89,441 deaths and left 4,65,979 persons injured during 2017 [**Table-1.6**].

The cause-wise analysis of deaths revealed that majority of deaths were due to traffic accidents which accounted for 45.1% (1,75,586 out of 3,89,441 deaths) of total deaths due to 'Other Causes' during the year 2017. Besides, 'Sudden Deaths', 'Drowning', 'Poisoning' and 'Falls' caused 11.0% (42,999 deaths), 7.8% (30,279 deaths), 5.3% (20,809 deaths) and 4.9% (19,092 deaths) of total deaths due to other causes respectively [**Table-1(C)**].

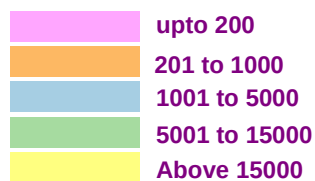
The age-group-wise distribution of accidental deaths in the country by various 'Other Causes' along with their share in total deaths is presented in **Table-1.7**. Majority of victims were in age groups 30 above - below 45 years and 18 above - below 30 years which accounted for 30.9% (1,20,319 deaths) and 27.1% (1,05,590 deaths) of total such deaths respectively.

State/UT-wise and gender-wise distribution of accidental deaths in the country by various other causes is presented in **Table-1.9**. Maharashtra with 9.2% (one-tenth) of country's population has reported 16.5% (64,159 out of 3,89,441 deaths) of accidental deaths due to other causes. Madhya Pradesh, Uttar Pradesh, Tamil Nadu, Rajasthan, Gujarat and Karnataka accounted for 10.3%, 8.7%, 7.2%, 7.1%, 6.2% and 6.0% of total such deaths respectively during the year 2017.

STATE/UT - WISE ACCIDENTAL DEATHS DUE TO OTHER CAUSES DURING 2017



Accidental Deaths



• As per data provided by States/UTs.

Map Powered by DevInfo, UNICEF

State/UT-wise and Cause-wise details on 'Other Cause of Accidents' is presented in **Table 1.9**. State/UT-wise analysis reveals that a total of 24,314 out of 1,75,586 deaths due to traffic accidents were reported in Uttar Pradesh accounting for 13.8% of total such deaths in the country during 2017. A total of 14,538 out of 42,999 deaths were reported from Maharashtra under 'Sudden Deaths' accounted for 33.8% of total such deaths. Large number of deaths due to 'Drowning' were reported from Madhya Pradesh (4,793 out of 30,279) accounting for 15.8% of such deaths. Maximum number of deaths due to 'Poisoning' were reported from Madhya Pradesh (4,294 out of 20,809) accounting for 20.6% of such deaths.

A total of 1,497 incidents of consumption of illicit/spurious liquor caused 1,510 deaths in the country during 2017. States where various such deaths were reported include Karnataka (256 deaths) followed by Madhya Pradesh (216 deaths), Andhra Pradesh (183 deaths), Punjab (170 deaths), Haryana (135 deaths), Puducherry (117 deaths) and Chhattisgarh (104 deaths) during 2017.

Fire Accidents

Considering the number of fire accidents in the country and potential of the damage that can be caused by such accidents, the Bureau has also collected data on fire accidents according to places of occurrence for appropriate preventive strategies viz. 'Fire in School Buildings', 'Fire in Commercial

Buildings', 'Fire in Residential / Dwelling Buildings', 'Fire in Government Buildings', 'Fire in Mines', 'Fire in Factories', 'Accidental Fire in Trains' and 'Accidental Fire in Vehicles' [**Table-1.10**].

A total of 13,397 cases of fire accidents were reported in the country during 2017, showing a decrease of 19.8% during 2017 over 2016 (16,695 cases). 13,397 incidents of fire accidents caused injuries to 348 persons and 13,159 deaths during 2017. The cause-wise analysis of fire accidents revealed that 57.9% of total deaths (7,614 out of 13,159) due to fire accidents were reported in residential/dwelling buildings during 2017 [**Table 1.11**].

Large number of cases of fire accidents were reported in Maharashtra (2,195 out of 13,397), accounting for 16.4% of total such cases during 2017 [**Table-1.11**].

The State/UT-wise analysis of place of fire accidents revealed that 21 States & UTs have reported 50.0% or more deaths due to 'Fire in residential or dwelling building' during 2017. These States/UTs were Mizoram, Tripura, A & N Islands (100% each), Jammu & Kashmir (93.3%), Meghalaya (92.3%), Punjab (87.3%), Gujarat (82.7%) and Kerala (80.8%) [**Table-1.11**].

Accidents in Cities

The term 'Mega City' refers to a city which has a population of 10 lakhs or more. As per the Population Census 2011, the status of mega city has been accorded to 53 cities.

A total of 61,023 accidental deaths were reported in 53 mega cities during 2017. Maximum numbers of accidental deaths (9,318 deaths) were reported in Mumbai accounting for 15.3% of total deaths, followed by Pune (4,982), Delhi City (4,450), Bengaluru (3,632), Jaipur (2,942) and Surat (2249).

Rate of accidental deaths in these 53 mega cities was higher (38.0) than all India rate of accidental deaths (30.3). The rate of accidental deaths was highest in Pune (98.7) followed by Aurangabad (96.9), Jaipur (95.8), Rajkot (92.9), Raipur (91.9) and Jodhpur (91.7) as compared to average rate for the cities (38.0) [Table-1.2].

Out of 61,023 accidental deaths in 53 mega cities; 460 and 60,563 deaths were due to forces of nature and other causes of accidents respectively during 2017. A decrease of 7.1% was observed in deaths due to forces of nature and 12.1% was witnessed in deaths due to 'Other Causes' during 2017 over 2016 respectively.[Table-1.3]

Kolkata, the third largest city in terms of population (8.8% of the total population of all mega cities) among 53 cities, reported the lowest share of deaths in cities accounting for 1.2% of total accidental deaths in 53 mega cities during 2017. The comparative reporting of 'Accidental Deaths' during 2017 in the top four-metropolitan cities is presented in Table-1(D).

Accidental Deaths due to Forces of Nature in Cities

A total of 460 deaths due to forces of nature were reported in these 53 mega cities during 2017. Among specified causes of accidents due to forces of nature in 53 cities, maximum deaths were reported due to 'Heat/Sun Stroke' followed by 'Exposure to cold', accounting for 30.2% and 27.4% of total such accidental deaths during 2017. All deaths in Vijayawada (31 deaths) were due to 'Heat/Sun Stroke' during 2017. Maximum deaths i.e. 100% in Chandigarh City (1 out of 1 death), 100% in Ghaziabad (4 out of 4 deaths), 75.0% (6 deaths out of 8 deaths) in Faridabad and 62.5% (20 deaths out of 32 deaths) in Ludhiana were due to 'Exposure to Cold'.

Accidental Deaths due to Other Causes in Cities

A total of 1,17,332 cases of 'Other' accidents were reported in 53 mega cities which caused injuries to 63,587 persons and 60,563 deaths during 2017. The cause-wise analysis of deaths due to other causes revealed that 27.9%, 20.7%, and 6.5% deaths in 53 cities were due to 'Traffic Accidents', 'Sudden Deaths' and 'Falls' respectively.

Among 53 cities, Mumbai city accounted for 15.4% of total deaths reported under 'Other' accidents followed by Pune and Delhi City which accounted for 8.2% and 7.3% respectively during 2017.

"Data on City-wise information is available on our website under 'Additional Tables for ADSI'"

Table – 1 (D)
Rate of Accidental Deaths in Metropolitan Cities during 2017

Sl. No.	City	Population (In Lakhs)	Share of Population In Cities (Percent)	Share of Deaths In Cities (Percent)	Rate (No. of Deaths Per One Lakh Population)
(1)	(2)	(3)	(4)	(5)	(6)
1.	Chennai	87.0	5.4	3.4	23.5
2.	Delhi	163.2	10.1	7.3	27.3
3.	Kolkata	141.1	8.8	1.2	5.2
4.	Mumbai	184.1	11.4	15.3	50.6

- As per data provided by States/UTs.
